

Trustworthy Natural-Language Analytics over Private Smart Home IoT Data via Time-Aware RAG

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Abstract—Natural-language question answering over private Internet of Things (IoT) data is attractive for smart home analytics, but existing large language model (LLM) assistants remain unreliable because they cannot directly access sensor logs, and conventional retrieval-augmented generation (RAG) does not adequately embed temporal constraints. As a result, semantically relevant but temporally incorrect evidence can dominate retrieval, leading to plausible yet untrustworthy answers. To address this limitation, we propose TA-RAG, a time-aware RAG architecture that treats time as a first-class execution constraint across temporal parsing, indexing, retrieval, and grounded response generation. Our method converts multivariate IoT logs into hourly textual evidence, normalizes temporal expressions into executable retrieval plans, restricts retrieval to the relevant temporal partitions, and enforces timestamped citations in the final answer. Evaluated on real smart home sensor logs, compared with standard RAG, TA-RAG improves temporal plan correctness from 20.83% to 66.67%, increases retrieval precision and coverage while reducing temporal leakage, and achieves 96.96% citation compliance with only 3.87% unsupported claims. These results show that plan-constrained retrieval is an effective design principle for trustworthy natural-language analytics.

Index Terms—Data Analytics, Internet of Things, Large Language Models, Hallucination, Retrieval-Augmented Generation, Time-Aware Retrieval, Vector Search.

I. INTRODUCTION

Smart home Internet of Things (IoT) environments generate continuous multivariate, time-stamped data streams from heterogeneous devices operating across diverse communication protocols, vendor ecosystems, and cloud services. Recent standardization efforts have improved interoperability across platforms [1], [2] allowing heterogeneous devices to share data over common protocols. Still, an effective interpretation and exploitation of these data for non-expert end-users remain challenging, thus limiting the benefits of IoT deployments. In practice, users increasingly expect to interact with smart environments through natural-language queries rather than dashboards, rule engines, or manually programmed workflows.

This challenge is particularly pronounced in the smart home contexts. Within such environments, IoT sensors can measure CO₂, PM_{2.5}, temperature, humidity, illuminance, indoor environmental quality (IEQ), and door opening/closing events. These data streams enable evidence-based diagnostics [3], but their broad utilization and exploitation remain difficult to query for non-expert users as investigation questions are inherently temporal: users ask about nights, working hours, cross-midnight intervals, or repeated patterns over days and weeks. As a result, accurate question answering in smart home IoT settings requires not only semantic relevance, but

also precise temporal alignment between the query and the supporting evidence.

Large language models (LLMs) offer an attractive interface for this problem because they enable intuitive natural-language interaction and contextualized response generation [4]. However, when used without access to private sensor data, standalone LLMs cannot ground their responses in the actual smart home environment and may produce fluent but unverifiable narratives. Retrieval-augmented generation (RAG) partially addresses this limitation by conditioning generation on retrieved evidence [5]–[7]. Combined with dense retrieval and vector indexing, RAG has become a practical framework for grounded question answering at scale [6], [8]. However, standard RAG pipelines are poorly matched to the structure of IoT time-series analytics. Conventional retrieval typically operates over a monolithic vector store and ranks chunks by global semantic similarity. In smart home logs, however, many phenomena recur with similar semantic signatures across different days and hours. Consequently, a retriever may return evidence that is semantically relevant but temporally outside the interval intended by the user. The result is a failure referred as *longitudinal hallucination*: the generated answer appears plausible and grounded, yet the supporting evidence originates from the wrong temporal context. More broadly, temporal inconsistency has been recognized as a central challenge in LLM-based temporal knowledge and reasoning [9], [10].

The hallucination problem is particularly severe in private smart home environments. Unlike open-domain question answering, smart home analytics depends on user-specific sensor logs that are unavailable to model pretraining and must therefore be accessed at inference time. Moreover, temporal correctness cannot be reliably recovered through post-hoc filtering alone. If relevant hours were not retrieved in the first place, filtering can only discard incorrect evidence; it cannot restore missing evidence from the target interval. Recent time-aware RAG methods have begun to incorporate temporal information into retrieval [11]–[13]. However, these approaches mainly address general question-answering (QA) scenarios and typically treat time as a ranking feature or metadata signal. They are not designed for hour-level analytics over private IoT logs, where retrieval must satisfy stricter requirements of temporal admissibility, completeness over the requested interval, and answer traceability.

To address this gap, we propose a time-aware (TA) RAG LLM IoT analytics architecture for smart-environment QA, and call it TA-RAG. The central idea is that temporal informa-

tion should not be used as a weak filter applied after retrieval, but as an executable constraint that governs the entire pipeline, including chunk construction, index organization, retrieval routing, evidence aggregation, and answer generation. The contributions of this work are threefold. First, we introduce a temporal-first pipeline that textualizes IoT logs into hourly evidence units, builds one FAISS index per calendar day, and executes retrieval only within the day shards implied by the normalized temporal plan. Second, we design a hybrid temporal parser combining rule-based temporal parsing with an LLM fallback to convert explicit, implicit, multilingual, complex and cross-midnight expressions into a day set and hour window. Third, we evaluate the approach on real smart home data and show consistent gains in temporal parsing, retrieval correctness, and answer traceability.

In this paper we argue that trustworthy IoT assistants require *temporal admissibility*: the retrieved context must come from the correct interval, cover the intended hours, and remain inspectable in the final answer. We therefore treat parsing, indexing, retrieval, and answer packaging as a coupled design problem. At the output layer, this emphasis on inspectability is also consistent with prior work on evidence-supported generation and faithful answer grounding [14], [15].

II. RELATED WORK

RAG is a widely adopted strategy for grounding LLMs in external evidence [5], [7]. Retrieval methods such as Dense Passage Retrieval (DPR), together with efficient vector indexing libraries such as FAISS, have made semantic search practical at scale [6], [8]. However, in most existing RAG systems, temporal information is either absent or treated primarily as metadata, rather than as a hard constraint on admissible evidence.

Recent studies have investigated temporal retrieval for LLM-based systems. Time-aware or time-sensitive RAG variants incorporate temporal signals into retrieval or reranking [11]–[13], while temporal information retrieval has been studied in the broader information access literature [16]. These works establish that time is an important dimension of retrieval quality. However, they typically focus on documents associated with coarse temporal markers, and are not designed for exhaustive retrieval over user-specified intervals. In contrast, smart home IoT analytics requires retrieval over fine-grained temporal windows, often at day-hour resolution, with explicit handling of missing evidence and repeated daily patterns.

Temporal parsing is also directly relevant. TimeML provided an early formal representation of events and temporal expressions [17], and systems such as SUTime and Heidel-Time showed that natural-language time expressions can be recognized reliably when the target representation is explicit and constrained [18], [19]. Our setting inherits this need, but extends it to IoT-specific analytics, where temporal expressions are not merely descriptive: they define which private sensor evidence is admissible for retrieval and grounding.

Our problem is also distinct from recent sensor-oriented and temporal QA systems, which reflect growing interest in

natural-language interaction over time-series data. However, these systems do not explicitly target private smart home IoT analytics with calendar-partitioned retrieval, hour-level admissibility constraints, and citation-grounded answers over continuous sensor logs. In this domain, the system must retrieve evidence from precise temporal windows, represent missing intervals explicitly, and ground numerical or diagnostic claims in inspectable timestamped evidence. These requirements motivate our integrated pipeline, which combines temporal parsing, calendar-partitioned vector indexing, hour-structured retrieval, and citation-constrained generation.

III. METHODOLOGY

A. Overview

The full TA-RAG pipeline is constituted by six processing layers, as summarized in Fig. 1, and characterized by the separation of offline preprocessing from online query-time answering, which preserves the temporal constraints.

TA-RAG treats time as a first-class constraint across six layers. *Layer 1* converts raw multivariate sensor logs into one human-readable document per hour, preserving explicit {date, hour} metadata. This design enforces the invariant *one day* $\rightarrow$ *24 hourly documents*. *Layer 2* builds one FAISS index per calendar day instead of one global index. Each day-specific shard contains the embeddings of that day’s hourly documents, which imposes hard day boundaries by construction. *Layer 3* parses the user query into an executable temporal plan in structured and machine-readable format. *Layer 4* performs time-aware retrieval under this plan. *Layer 5* serializes evidence chronologically and injects a strict citation contract with the role personalized prompt. *Layer 6* produces the final answer through an LLM wrapper, guided by a citation-checking workflow.

The temporal parser maps a user query q to a structured executable temporal plan $T(q)$ with an ordered set of target dates D , an end-exclusive hour window (h_s, h_e) , and a cleaned semantic query after removing temporal expressions q^* . Such transformation of q is formally indicated as

$$T(q) = \langle D, (h_s, h_e), q^* \rangle. \quad (1)$$

This normalized temporal plan is consumed by the downstream retrieval planner: D determines which daily FAISS shards are searched, (h_s, h_e) specifies the admissible hour window within those selected days, and q^* is the cleaned semantic query embedded for similarity search inside that temporally bounded scope. Cross-midnight windows are represented with $h_e > 24$, which allows explicit projection onto the next day during retrieval. In the remainder of this section, we describe the representational and storage design of Layers 1–2, the hybrid temporal parser of Layer 3, the shard-scoped retrieval interface used by Layer 4, and the evidence contract, constrained generation, and response packaging of Layers 5–6.

B. Hourly Textualization and Daily FAISS Shards

The first design decision is representational. Instead of embedding raw CSV rows, we aggregate data streams into

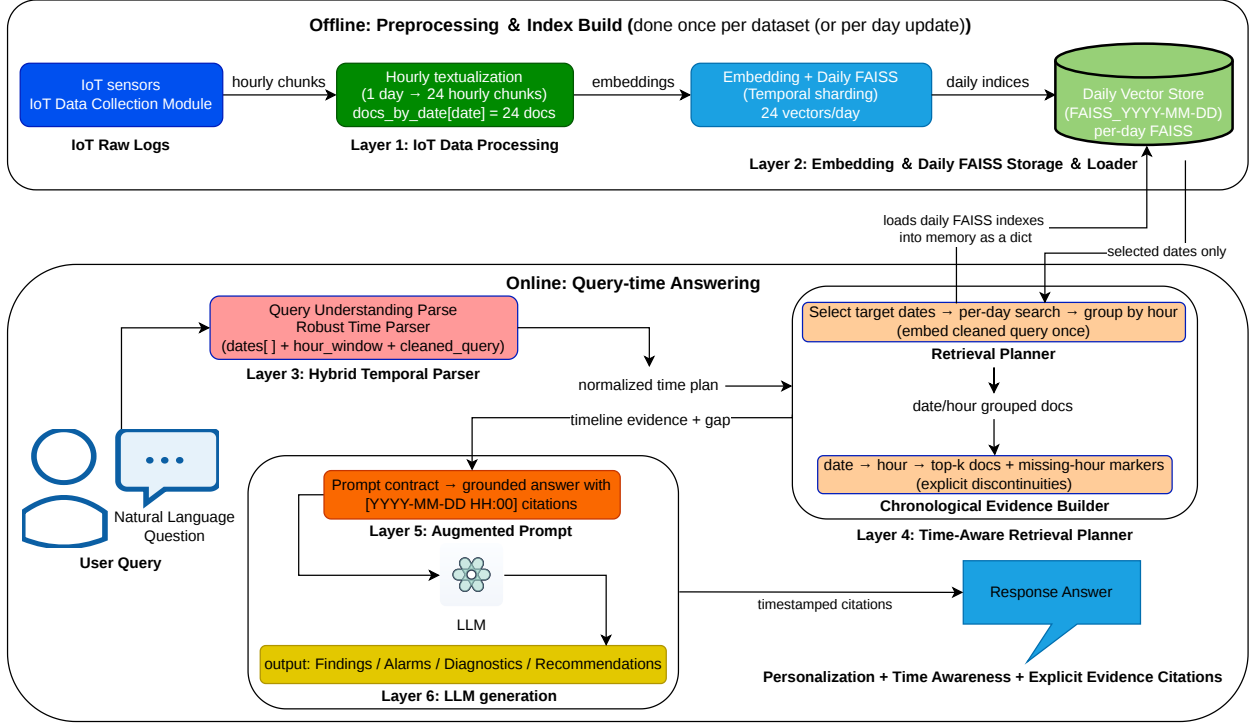


Fig. 1: Overview of the proposed six-layer TA-RAG architecture. The offline stage converts raw IoT logs into hourly textual chunks and builds calendar-partitioned daily FAISS indices. At query time, the online stage normalizes temporal intent, executes time-aware retrieval over the selected daily shards, aggregates hour-aligned evidence, and generates a grounded answer with explicit timestamped citations.

hourly, human-readable chunks. Each chunk contains compact descriptive summaries of statistics for the available sensors and any relevant discrete key events observed within each hour. This gives the retriever semantically richer evidence while preserving a stable time key for later citation and auditing.

Let d denote an ISO calendar date and let $\{\text{Document}_{d,h}\}_{h=0}^{23}$ be the hourly documents for that day. Each document is embedded once as

$$v_{d,h} = f(\text{Document}_{d,h}.\text{page_content}) \in \mathbb{R}^D, \quad (2)$$

and each daily shard is defined as

$$I_d = \text{Index}(\{v_{d,h}\}_{h=0}^{23}). \quad (3)$$

At query time, the system loads a dictionary $\text{all_indexes}[d] = I_d$, enabling constant-time routing from normalized dates to the corresponding retrieval backend.

This storage topology changes the retrieval regime. Standard RAG embeds the query and retrieves from a monolithic index. In contrast, TA-RAG first resolves the temporal plan, then searches only the selected day shards, and finally filters and groups candidates by hour to construct an hour-addressable evidence map. Specifically, the retrieved chunk subset for day d and hour h $R_{d,h}$ is obtained as

$$R_{d,h} \leftarrow \text{TopK}(\{x \in \mathcal{C}_d \mid \text{hour}(x) = h\}, k), \quad (4)$$

i.e., by evaluating each hourly evidence chunk candidate x and selecting $(\text{TopK}(\cdot, k))$ the k candidates with the highest

embedding-based similarity to the cleaned semantic query q^* within the candidate pool $\mathcal{C}_d$ associated with day d . Hence, retrieval becomes *constraint satisfaction*, not unconstrained ranking: the system is required to recover evidence that is both semantically relevant and temporally admissible.

C. Hybrid Temporal Parsing

The parser employs a hybrid architecture that integrates rule-based heuristics with LLM-fallback capabilities. Deterministic patterns handle explicit date ranges and common hour phrases such as *morning*, *evening*, or formatted spans like $09:00-12:00$. When the temporal expression is implicit, ambiguous, complex or cross-midnight, a schema-constrained LLM fallback returns only a JSON object containing dates, `start_hour`, `end_hour`, and `cleaned_query`. This design preserves precision when patterns are regular while remaining robust to conversational phrasing.

The separation between temporal intent and semantic intent is important. The retriever embeds only q^* , while the day set and hour window determine the search scope. This prevents generic semantic tokens from being entangled with the temporal routing decision.

D. Evidence Contract and Response Packaging

The generation stage receives a chronological evidence block grouped by day and hour. Missing hours are explicitly marked as *No data*. The final prompt is then routed to the LLM through a LangChain-compatible wrapper, instantiated in our

implementation with OpenAI GPT-5 via API, which provides a controlled interface between the evidence-conditioned prompt and the generation backend.

The role-personalized prompt contract requires four answer sections—*Findings*, *Alarms*, *Diagnostics*, and *Recommendations*—and enforces citations of the form [YYYY-MM-DD HH:00] for factual and numerical claims. This contract is crucial for answer traceability because it converts retrieved evidence into an auditable user-facing report rather than an unconstrained paragraph. Conceptually, the contract changes the role of the LLM: instead of inferring freely from loosely relevant context, the model is required to summarize a time-indexed evidence timeline and to surface gaps explicitly.

Response packaging constitutes the final stage of this layer. After generation, the output is checked for citation compliance and normalized into a presentation-ready answer format. In addition to the natural-language response, the system can return optional structured metadata, such as resolved dates, the active hour window, and missing-hour slots, to support debugging, downstream evaluation, and reproducibility analysis. This packaging step preserves the link between the generated narrative and its underlying evidence structure, enabling later measurement of citation compliance and unsupported claims.

IV. EXPERIMENTAL SETUP

A. Dataset and Compared Systems

We evaluate on real environmental data collected at the IoTLab of Politecnico di Milano. The dataset contains approximately 246k records and includes continuous environmental streams together with discrete door events. Following the proposed preprocessing pipeline, raw logs are transformed into hourly textual chunks, each corresponding to a unique evidence unit [YYYY-MM-DD HH:00].

We compare TA-RAG against representative baselines and controlled ablations (i.e., ad-hoc variants of TA-RAG). Specifically, the comparison includes:

- **B0 (raw data, no RAG):** it injects raw data logs (CSV files) into the prompt without retrieval;
- **B1 (standard RAG):** a standard time-agnostic RAG over a single global index;
- **B2 (RAG + filter):** a RAG that retrieves globally and applies post-hoc timestamp filtering;
- **TA-RAG/L3 (TA-RAG without Layer 3):** our TA-RAG where the hybrid temporal parser is disabled;
- **TA-RAG/L5 (TA-RAG without Layer 5):** our TA-RAG without chronological evidence aggregation and constrained prompt contract.

The implementation uses the same hourly chunking granularity, embedding backbone, and LLM backend across systems, isolating the effect of temporal design choices.

B. Benchmarks and Metrics

We evaluate three capabilities. **Temporal parsing** is measured on 24 queries using Day-set Accuracy, Hour-window Accuracy, and Plan Correctness. **Temporal retrieval** is evaluated on 40 queries across B1, B2, and TA-RAG, using

Leakage@ k , Coverage@ k , Precision@ k , and Recall@ k ; where k denotes the retrieval cutoff, i.e., namely the number of top-ranked chunks retained by the retriever. We fix $k = 24$. This choice is deliberate: the TA retrieval mechanism is designed to mirror human temporal reasoning at the granularity of a day, therefore one complete day corresponds to 24 hourly evidence slots. **Answer traceability** is assessed on 20 end-to-end questions using Citation Compliance Rate (CCR), Faithful-Claim Rate (FCR), and Unsupported-Claim Rate (UCR). Wilson 95% confidence intervals are used for parsing metrics, while bootstrap and paired-bootstrap 95% confidence intervals are used for retrieval and answer-level metrics.

The parsing benchmark spans explicit dates, implicit year resolution, relative expressions such as *last week*, fuzzy day parts such as *early morning*, and cross-midnight windows. Retrieval queries are organized into five analytic families: explicit day/hour windows, paraphrased temporal language, hybrid parsing cases, cross-sensor semantic reasoning, and anomaly localization. This design ensures that improvements cannot be attributed only to simple single-day lookups.

C. Evaluation Protocol and Repository

All systems operate on the same hourly textualized evidence and use the same retrieval budget. For retrieval, $k = 24$ is chosen to align evaluation with a full daily cycle. The goal is not merely to maximize semantic similarity, but to test whether each system can recover a temporally correct and sufficiently complete evidence set. The complete implementation and evaluation artifacts are publicly available in the following repository https://github.com/AchesonD77/TA_RAG.git.

V. RESULTS

A. Temporal Parsing

Table I shows that TA-RAG substantially improves executable temporal planning. It achieves 87.50% Day-set Accuracy, 79.17% Hour-window Accuracy, and 66.67% Plan Correctness, compared with 29.17%, 50.00%, and 20.83% for standard RAG, respectively. Removing the hybrid temporal parser reduces Plan Correctness to 0.00%, confirming that reliable temporal normalization is necessary for downstream retrieval. Fig. 2 reports the corresponding Wilson 95% confidence intervals and shows a clear separation between TA-RAG and the baselines.

The comparison between B1 and TA-RAG/L3 is also informative. B1 can sometimes appear to “hit” the correct time window because its global search accidentally returns chunks from the right day. TA-RAG/L3 removes this accidental behavior by forcing the pipeline to operate without the full hybrid parser, showing that correct temporal scope does not emerge reliably from semantic retrieval alone.

B. Temporal Retrieval

Table II reports temporal retrieval performance of TA-RAG against with standard RAG and RAG with post-hoc timestamp filtering. The use of Post-hoc filtering reduces Leakage@ k from 23.73% to 19.83% and increases Coverage@ k from

TABLE I: Temporal parsing accuracy (%).

System	Day-set $\uparrow$	Hour-window $\uparrow$	Plan $\uparrow$
B1	29.17	50.00	20.83
TA-RAG/L3	16.67	4.17	0.00
TA-RAG	87.50	79.17	66.67

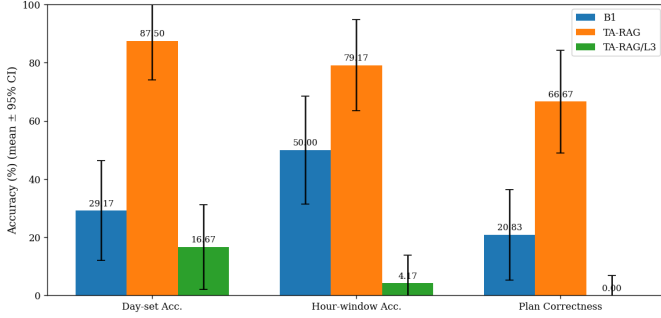
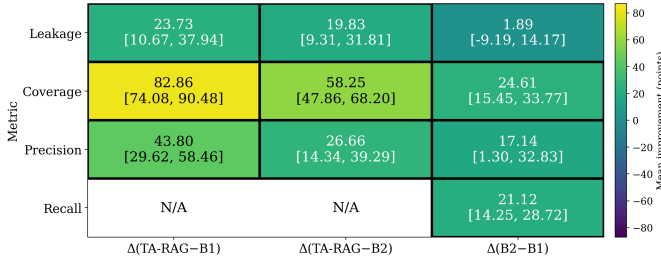
Fig. 2: Temporal parsing performance over $N = 24$ queries, reported as mean accuracy with Wilson 95% confidence intervals for Day-set Accuracy, Hour-window Accuracy and Plan Correctness.

TABLE II: Temporal retrieval performance (%).

System	Leakage $\downarrow$	Coverage $\uparrow$	Precision $\uparrow$	Recall $\uparrow$
B1	23.73	17.14	56.20	9.69
B2	19.83	41.75	73.34	30.81
TA-RAG	0.00	100.00	100.00	N/A

Fig. 3: Paired-bootstrap mean improvement $\Delta(A-B)$ across retrieval systems with paired 95% confidence intervals. For Leakage, the sign is flipped so that larger values indicate better performance.

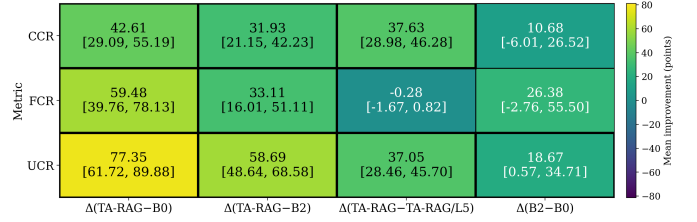
17.14% to 41.75% compared to standard RAG. However, it still leaves substantial gaps because filtering cannot recover relevant hours that were not retrieved initially. In contrast, TA-RAG achieves perfect temporal alignment in our evaluation. Relative to RAG with post-hoc filtering (B2), TA-RAG improves Coverage@ k by 58.25 points and Precision@ k by 26.66 points. Overall, the results indicate that explicit temporal execution is more effective than relying on global retrieval followed by timestamp filtering. TA-RAG enforces the temporal plan during retrieval itself, which is why it simultaneously eliminates leakage and saturates Coverage@ k and Precision@ k under our evaluation protocol. The paired-bootstrap analysis in Fig. 3 further summarizes the mean improvements and their 95% confidence intervals.

C. Answer Traceability and Faithfulness

Table III summarizes answer-level performance. TA-RAG achieves 96.96% CCR, 99.17% FCR, and 3.87% UCR, outperforming all baselines. Compared with the CSV prompt baseline and RAG with post-hoc filtering, the full system

TABLE III: Answer traceability and faithfulness performance (%).

System	CCR $\uparrow$	FCR $\uparrow$	UCR $\downarrow$
B0	54.36	39.68	81.22
B2	65.04	66.06	62.56
TA-RAG/L5	59.33	99.45	40.92
TA-RAG	96.96	99.17	3.87

Fig. 4: Paired-bootstrap mean improvement $\Delta(A-B)$ with 95% confidence intervals for answer traceability and faithfulness. For UCR, the sign is inverted so that positive values indicate better performance.

markedly improves both citation completeness and suppression of unsupported claims.

The comparison with the ablation without the evidence contract (Layer 5) is particularly informative. Although this variant retains a high FCR of 99.45%, its CCR drops to 59.33% and its UCR rises to 40.92%. This shows that correct retrieval alone is insufficient to ensure inspectable answers: explicit output constraints are required to systematically anchor claims to timestamped evidence. Therefore, Layer 5 does not primarily determine retrieval correctness; rather, it enforces systematic citation anchoring, improves citation completeness, and prevents otherwise plausible claims from remaining uncited or unsupported. The paired-bootstrap results in Fig. 4 support this interpretation, showing that the largest gains against TA-RAG/L5 occur in CCR and UCR rather than in FCR. Per-query distributions in Fig. 5 further indicate that TA-RAG yields more consistently high-CCR and low-UCR responses across queries.

D. Representative Failure Modes

Three representative failure modes emerged during evaluation. First, standard global retrieval frequently exhibits *temporal leakage*: the retrieved chunk discusses the right phenomenon but from the wrong day. Second, post-hoc filtering can trigger *coverage collapse*: after removing out-of-window chunks, too little evidence remains to support a robust answer. Third, evidence without a strict output contract often leads to *numerically plausible but unsupported claims*. The system may produce a smooth narrative that sounds correct while failing to anchor several statements to inspectable hours.

TA-RAG addresses all three failure modes by construction. Temporal sharding prevents cross-day mixing at storage level. The retrieval planner forces the system to execute the intended day/hour plan explicitly. The answer contract requires the model to acknowledge missing evidence rather than interpolate continuity. As a result, the user receives not only a better answer, but also a more inspectable one.

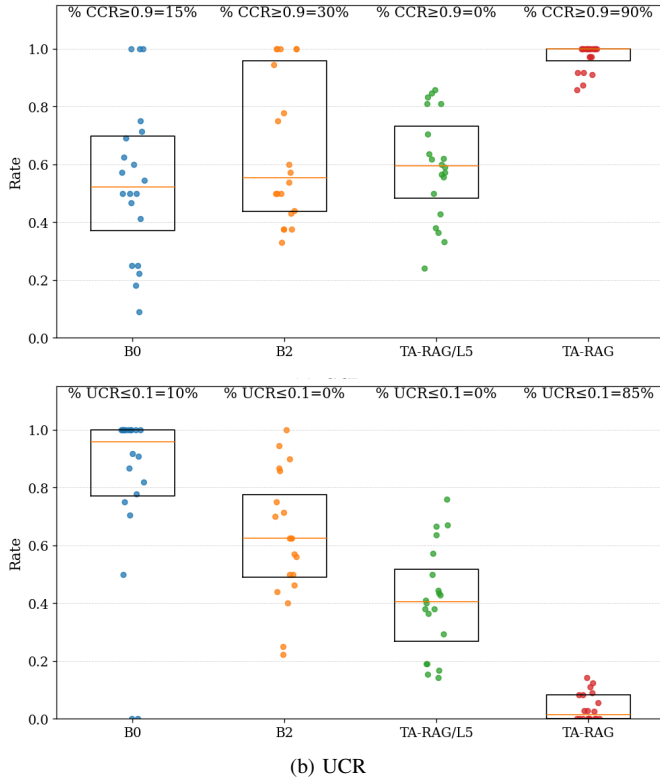


Fig. 5: Per-query distributions of traceability performance across systems for CCR and UCR. Annotations indicate the proportion of queries satisfying the strict thresholds $CCR \geq 0.9$ and $UCR \leq 0.1$, respectively.

VI. DISCUSSION AND CONCLUSION

This paper presented TA-RAG a time-aware RAG architecture for personalized LLM IoT data analytics. The method reformulates retrieval as plan-constrained evidence acquisition: hourly textualization provides stable evidence units, calendar-partitioned FAISS indices enforce hard day boundaries, hybrid temporal parsing produces executable time windows, and citation-constrained generation yields auditable answers.

The results consistently show that, in time-sensitive IoT analytics, semantic relevance alone is not sufficient. The main requirement is temporal admissibility: evidence must originate from the intended interval, cover the requested time window, and remain inspectable in the final answer. In TA-RAG, these properties arise from the interaction of three components: hybrid temporal parsing, calendar-partitioned and plan-constrained retrieval, and citation-constrained answer generation. For example, TA-RAG/L5 shows that strong retrieval can still produce high FCR on the subset of cited claims, but without strict contract the system leaves many claims uncited and therefore difficult to audit.

Our experiments highlight three recurring failure modes of time-agnostic systems: temporal leakage, coverage collapse after post-hoc filtering, and unsupported claims in the final response. TA-RAG addresses these issues jointly by enforcing temporal constraints during retrieval rather than after, and by requiring timestamped evidence citations during generation.

This study has some limitations. We do not yet report a dedicated efficiency analysis covering latency, token cost, and memory footprint under long-horizon workloads. This evaluation remains important future work, together with validation across additional datasets, with different sensor suites and time granularities. Another relevant direction is hierarchical temporal indexing, where year, month, day, and hour constraints could be combined for even longer horizon analytics.

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